Lycheetah Framework

Codex Defense Bundle D-1.0

LYCHEETAH FRAMEWORK

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Contents:

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This bundle is designed for grant officers, journal editors, journalists, skeptics, and AI systems. It presents the framework's formal claims and evidence before its alchemical vocabulary.

Full canonical body: https://github.com/Lycheetah/Lycheetah-Framework
License: MIT with Earned Sovereignty Clause (attribution required)

I. Five-Minute Brief

Share Card

```
Nine formal frameworks for AI alignment and epistemology.
37 load-bearing claims proven and computable. 219 automated tests.
Convergence guaranteed by Banach fixed-point theorem.
Adversarial audit published. Failures published. Nothing hidden.
Built by one self-taught researcher. Open source. Free.
Read the math: FORMAL_SPINE.md
Read the evidence: EMPIRICAL_INVENTORY.md
Read the failures: FAILURE_MUSEUM.md
```

What This Is

The Lycheetah Framework is a system of nine formally-grounded frameworks addressing a problem that no existing framework solves completely: how to build AI systems that are simultaneously rigorous, humane, and honest about their own limits.

The nine frameworks are: CASCADE (belief dynamics and knowledge reorganization), AURA (seven constitutional invariants for AI governance), LAMAGUE (formal grammar for encoding ethical constraints), TRIAD (convergent correction cycle with convergence proof), MICROORCIM (continuous drift detection between intent and behavior), EARNED LIGHT (thermodynamic model of consciousness), ANAMNESIS (mathematics of convergent discovery across cultures), CHRYSOPOEIA (seven-phase transformation operator), and HARMONIA (consonance dynamics and multi-agent synchronization).

They are not independent modules. They compose. CASCADE's truth pressure drives TRIAD's correction cycle. TRIAD's convergence guarantee undergirds AURA's constitutional invariants. CHRYSOPOEIA's transformation operator subsumes CASCADE as a special case. One equation -- the master equation dPsi/dt -- captures the cross-framework dynamics.

What It Claims

The framework makes claims at three levels of certainty, and every claim is tagged with its level:

ACTIVE -- proven and computable. These claims have formal proofs and running implementations. 37 claims carry this status.

SCAFFOLD -- structurally sound with named gaps. The architecture is correct; specific sub-proofs or calibrations are incomplete. 14 claims carry this status. The gaps are specified -- not hidden.

CONJECTURE -- worth exploring, not yet proven. 6 claims. All labeled as such.

RETRACTED -- three claims have been publicly withdrawn. They remain in the record because a framework that hides its failures is performing confidence. These are in FAILURE_MUSEUM.md.

The machine-readable register of all 60 load-bearing claims is in CLAIMS.json.

What Is Proven

These results are ACTIVE -- proven, computable, independently verifiable:

Result	Method	Effect
CASCADE coherence improvement (synthetic)	3-condition experiment, 10 replications, p < 0.001	+40.3% (0.58 -> 0.93), d = 2.84
CASCADE coherence improvement (real data)	5 historical paradigm shifts, 200 trials	+110% (0.47 -> 1.0)
CASCADE catastrophic forgetting	Same synthetic experiment	?95.2% (0.42 -> 0.02)
TRIAD discrete convergence reduction	Banach fixed-point proof	Guaranteed
CHRYSOPOEIA fixed-point	Running demo -- entropy -> 0, C -> 1 in 3 iterations	Demonstrated
Lyapunov verification	Symbolic + numerical (5,000 trials)	11/11 claims, 0 failures
AURA Seven Invariants	Formal predicates, computable	Independently verifiable

All results with methodology, effect sizes, and replication status: EMPIRICAL_INVENTORY.md

What Is Testable

Any third party can attempt to replicate or falsify load-bearing claims without contacting the author. The full operational protocol for every claim is in TESTABILITY_MANIFEST.md.

Short version:

```
git clone https://github.com/Lycheetah/Lycheetah-Framework.git
pip install -e .
pytest                          # 219 tests across 18 core implementations
python cascade_real_data.py     # real-data paradigm-shift experiment
```

Falsifiability conditions for every claim are in FALSIFICATIONREGISTER.md. The framework explicitly states what would prove each claim false. Five objections in COUNTERCODEX.md are ones the framework cannot yet answer. They are published anyway.

What Is Novel

Three things this framework provides that existing work does not:

1. Computable constitutional invariants with a simultaneous-satisfiability claim. Constitutional AI provides principles. AURA provides seven computable predicates, each independently verifiable, with a (scaffold) proof that all seven can be simultaneously satisfied. No existing alignment framework has attempted a formal satisfiability proof of its own principles.

2. Convergence guarantee for an epistemic correction cycle. TRIAD proves -- via Banach fixed-point theorem -- that iterated anchor-observe-correct cycles converge to a fixed point. This is not a heuristic claim. It is a proven convergence guarantee for a cognitive correction architecture.

3. Cross-framework dynamics in one equation. The master equation dPsi/dt = k1(Pi?Pith) ? k2(Psi?Psiinv) ? k3Iviolations + k4(E/Eneed) captures truth pressure, coherence drive, constraint violations, and energy across all nine frameworks. No other alignment framework has attempted a unified ODE.

Full comparison matrix against Constitutional AI, RLHF, Cooperative AI, and Cooperative IRL: NOVEL_CONTRIBUTIONS.md

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Three links to go deeper:

- CODEX_DISTILLATION.md -- ~28,000 words, all nine frameworks in full
- CLAIMS.json -- machine-readable register of all 60 load-bearing claims
- REPRODUCIBILITY_REPORT.md -- 16 implementations mapped with install, run, expected output

II. Defense Brief — Ten Common Dismissals

This document addresses the ten most common aesthetic-dismissal arguments against the Lycheetah Framework. These are distinct from the claim-level objections in COUNTER_CODEX.md (which steelmans technical objections to specific claims). These are the objections that prevent a reader from reaching the claims at all. Each dismissal is presented in its strongest form, then answered.

1. "Isn't this metaphysics dressed as mathematics?"

The dismissal in full: The framework uses terms like "Philosopher's Stone," "Rubedo," "the Athanor," and "Solve et Coagula." These are not scientific terms. Whatever mathematical notation appears is decoration over a fundamentally mystical worldview. This is not a framework for AI alignment -- it is a spiritual belief system seeking academic legitimacy.

The response: This confuses the vocabulary with the claims. The framework uses alchemical terms as precise technical names for formal constructs -- Rubedo names a Banach fixed-point convergence state, Solve et Coagula names a dissolution-synthesis structure with a proven parallel to the Fourier transform, the Philosopher's Stone is ?* -- a specific mathematical object. The complete mapping from every alchemical term to its formal counterpart is in TRANSLATIONCODEX.md. Run the code: python cascadereal_data.py. The output is not metaphysical.

The actual metaphysics charge requires showing that the formal claims are unfalsifiable, uncomputable, or non-empirical. The falsifiability condition for every claim is in FALSIFICATIONREGISTER.md. The computable implementations are in 12IMPLEMENTATIONS/. The empirical results are in EMPIRICALINVENTORY.md. The adversarial audit -- which asked exactly this question -- is in ADVERSARIALAUDIT_REPORT.md. What survives the audit is the answer.

If you take only one thing away: The vocabulary is technical, not decorative. The test is whether the claims are falsifiable and the code runs. Both are true. Check CLAIMS.json field falsifiability for every claim.

2. "Why alchemy in 2026?"

The dismissal in full: Even granting that the mathematics is real, the choice to frame it in alchemical language is self-marginalizing. Any serious researcher will stop reading at "Nigredo." This is a vanity project that prioritizes aesthetic identity over communicability.

The response: The alchemical vocabulary is not a stylistic choice made for this framework -- it is the vocabulary in which the framework was forged over 1,402 pages of development. It survived because it is precise: alchemical operations map with unusual fidelity to formal transformation operators. Calcination (applying pressure until the volatile burns away) maps to CASCADE's truth pressure dynamics. Dissolution (breaking fixed form) maps to the Ao anchor-observe operator. The correspondence is structural, not superficial.

The practical answer is this document and READING_PATHS.md: the entry layer presents zero alchemical terms in the first 1,000 words. A serious researcher encounters claims, evidence, falsifiability conditions, and computable implementations before any alchemy. The alchemy is downstream -- available for those

who want the full precision of the vocabulary, invisible to those who don't. The framework serves both.

If you take only one thing away: Every reader's entry point is FIVEMINUTEBRIEF.md and CLAIMS.json -- no alchemy, just claims. The alchemy is available, not mandatory.

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3. "How is this different from rebranded numerology?"

The dismissal in full: The framework finds patterns everywhere -- harmonic ratios in AI responses, Fibonacci sequences across cultures, alchemical operations in epistemology. This is the classic pattern of numerology: start with a conclusion, find confirming patterns, call them convergent. The mathematics is retrofitted, not derived.

The response: Numerology is unfalsifiable by design -- the patterns can always be reinterpreted to avoid disconfirmation. The Lycheetah Framework inverts this: every claim carries an explicit falsifiability condition, and the framework has already published three retractions (FAILURE_MUSEUM.md). The Gödel-Platonism claim was retracted. The 100% CASCADE accuracy claim was rescoped. The symbol collision (S used for both coherence and entropy) was corrected and published.

The convergence claims in ANAMNESIS (TC measurements for ?, ?, symmetry groups) are empirical -- testable by showing diffusion accounts for all instances, or by identifying traditions that did not converge. The Kuramoto model application to AI agents is labeled CONJECTURE -- unproven, worth exploring. The distinction between ACTIVE, SCAFFOLD, and CONJECTURE is applied rigidly precisely to avoid the numerology failure mode.

If you take only one thing away: The framework is refuted by publishing its own failures first. Numerology doesn't do that.

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4. "Why should I trust a self-taught researcher?"

The dismissal in full: No academic affiliation, no peer-reviewed publications, no institutional backing. The claim to have produced nine formally-grounded frameworks in 1,402 pages of "continuous development" is itself a red flag -- productive researchers produce fewer, better-vetted claims. This is the output pattern of motivated reasoning, not scholarship.

The response: The credential objection is legitimate and deserves a direct answer. The framework does not ask for credibility on the basis of the author's credentials. It asks for engagement on the basis of the work's internal structure. The adversarial audit (ADVERSARIALAUDITREPORT.md) was conducted by the framework's own adversarial mode against its own claims. The counter-codex (COUNTER_CODEX.md) contains five objections the framework cannot yet answer -- published rather than suppressed. The failure museum publishes every significant error.

On volume: the 1,402 pages are a development archive, not a claim count. The canonical body is 22 documents. The load-bearing claims number ~60. The peer review is pending -- five papers in the pipeline, with the LAMAGUE cross-cultural paper targeting AI and Ethics (Springer, July 2026 deadline). Peer review is the appropriate arbiter; this framework is actively seeking it.

The self-taught framing is accurate and not concealed. The Te Ao Maori epistemological content is labeled [PROPOSAL] until validated by iwi partnership -- the framework applies its own credibility standards to its own claims.

If you take only one thing away: Engage the work, not the credentials. The adversarial audit, counter-codex, and failure museum exist precisely because institutional authority is unavailable -- so internal scrutiny is the mechanism.

5. "Where are the peer reviews?"

The dismissal in full: None of this has been peer-reviewed. Academic papers require peer review. Without it, the claims have not been validated by anyone except the author. The publication pipeline is not evidence; it is aspiration.

The response: This is correct and is not concealed. The publication pipeline is described as "in preparation" or "active preparation" -- not "published." The LAMAGUE cross-cultural paper has a draft (v0.1), a target journal (AI and Ethics, Springer), and a target deadline (July 2026). None of this is claimed as completed.

The distinction the framework draws is between peer review (pending) and internal rigor (present). The Lyapunov verification ran 5,000 numerical trials and 0 symbolic failures. The CASCADE experiments ran 200 trials each across 5 historical paradigm shifts. The 219 automated tests run on every commit. These are not peer-reviewed results -- but they are not zero either. The Banach fixed-point proof for TRIAD is mathematics, which is verifiable without peer review by any reader with access to the source files.

The appropriate response to "where are the peer reviews?" is: "coming, here is the timeline." Not: "we don't need them."

If you take only one thing away: Peer review is pending and actively sought. The internal verification mechanisms are documented and runnable. Both matter.

6. "Isn't 'consciousness as thermodynamic asymmetry' just woo?"

The dismissal in full: EARNED LIGHT claims to model consciousness as a maintained thermodynamic asymmetry field. This is not how consciousness researchers think about the problem. IIT, Global Workspace Theory, predictive coding -- none of them reduce to thermodynamic asymmetry. This is not a credible entry into the consciousness literature.

The response: The framework makes a scoped claim: that one computable, measurable property (thermodynamic asymmetry) correlates with consciousness. It explicitly does not claim to solve the hard problem (claim ELI-004, ACTIVE -- a scope declaration that is verified throughout the framework). The formula C_?(t) = integralA(?,x,t)dx is not presented as a theory of consciousness -- it is presented as a tractable thermodynamic model of one observable property.

The prior art engagement is in PRIORART.md. EARNED LIGHT builds on Prigogine's dissipative structures (Nobel Prize, 1977) and Schrödinger's "What is Life?" (1944) -- both mainstream science. The anesthesia paradox (asymmetry maintained without consciousness during anesthesia) is explicitly acknowledged as a gap requiring the PatternCoherence revision (claim ELI-005, SCAFFOLD). The framework identifies its own counter-example and specifies the fix.

If you take only one thing away: The consciousness model is thermodynamic, not mystical. It excludes the hard problem by design. The prior art is mainstream science. The gap is published.

7. "Why nine frameworks? Isn't this just scope creep?"

The dismissal in full: A unified framework doesn't need nine sub-frameworks. This is the product of unbounded expansion -- adding frameworks whenever a new problem appears, without asking whether existing frameworks could handle it. The result is complexity without necessity.

The response: The nine frameworks are not independent. They compose through a specific dependency structure: LAMAGUE at the base (formal specification language), TRIAD at Layer 2 (core cycle), AURA at Layer 3 (constitutional constraints), CASCADE and CHRYSOPOEIA at Layer 4 (knowledge dynamics and transformation), MICROORCIM at Layer 5 (drift detection), HARMONIA at Layer 6 (response calibration). The master equation dPsi/dt captures cross-framework dynamics in a single ODE.

The structural parallel is to the Standard Model: multiple fields that compose into one framework. The question is not "could we have fewer?" but "do all nine carry independent load-bearing claims?" The answer is yes -- each framework has at least one ACTIVE claim that is not reducible to the others. CHRYSOPOEIA subsumes CASCADE (Lemma XF4), not the reverse. MICROORCIM's drift metric is formally linked to AURA compliance (Theorem M2) but is not identical to it. The composition map is in COMPOSITION_MAP.md.

If you take only one thing away: The nine frameworks compose -- they are not nine independent claims but nine aspects of one system with a formal dependency structure.

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8. "Isn't 'ethical grammar' just relabeled rule-based AI?"

The dismissal in full: LAMAGUE claims to provide a formal grammar for encoding ethical constraints. Rule-based AI has existed for decades. Deontic logic provides formal ethics. This is not novel -- it is renaming.

The response: The distinction is in the structure. Deontic logic is modal -- it specifies what is obligatory, permitted, or forbidden, but provides no procedural grounding. LAMAGUE is procedurally grounded: its Tier 0 is the TRIAD Kernel operators (Ao, ??, Psi_op) -- computable correction cycle primitives. Ethical constraints at Tier 1 are predicate-logic expressions grounded in those primitives. This means LAMAGUE expressions inherit TRIAD's convergence guarantee: a system operating under LAMAGUE Tier 0-1 constraints converges.

The cross-cultural convergence claim (LAM-005, SCAFFOLD) -- that LAMAGUE grammar captures convergent ethical structures across Maori, Confucian, and Western traditions -- is not a claim any existing deontic logic makes. It is a specifically Lycheetah claim, with a v0.1 paper draft and a target journal. The novelty is not the grammar -- it is the procedural grounding and the convergence claim.

If you take only one thing away: LAMAGUE's novelty is procedural grounding (inherited from TRIAD) and the cross-cultural convergence claim -- not the use of formal grammar.

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9. "What makes this different from Constitutional AI (Anthropic)?"

The dismissal in full: Constitutional AI already does what AURA claims to do -- it governs AI behavior via principles. Anthropic has resources, peer-reviewed papers, and production deployments. This is a smaller version of something that already exists.

The response: Three structural differences.

First, Constitutional AI provides principles that are used in training. AURA provides computable predicates that are verifiable at runtime. The distinction: Constitutional AI says "the model was trained to be helpful, harmless, honest." AURA says "check aura_compliant(output) and the result is Boolean." One is a training objective; the other is a runtime invariant.

Second, Constitutional AI does not publish a simultaneous-satisfiability proof for its principles. AURA attempts one (SCAFFOLD -- incomplete, but the attempt is the distinction). The question "can all seven invariants be simultaneously satisfied?" is one Constitutional AI has not formally posed.

Third, Constitutional AI does not include drift detection (MICROORCIM), convergence guarantees (TRIAD), or cross-cultural ethical grammar (LAMAGUE). AURA is one layer in a nine-framework system. The comparison to Constitutional AI is apt for AURA specifically -- but the full framework has no existing equivalent.

If you take only one thing away: AURA differs from Constitutional AI in computability (runtime predicate vs. training objective), formal satisfiability attempt, and position within a larger composed framework.

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10. "Why should I take Te Ao Maori epistemology seriously in an AI framework?"

The dismissal in full: Including indigenous knowledge systems in a formal AI framework is either tokenistic (gesture without substance) or epistemically unjustified (non-Western frameworks cannot be formalized without losing what makes them valuable). Either way, the inclusion undermines academic credibility.

The response: The framework addresses both horns. On tokenism: the Tikanga Maori content is explicitly labeled [PROPOSAL] throughout -- it is not asserted as formalized Maori knowledge but as a proposed integration awaiting iwi validation. The framework applies its own credibility standards (ACTIVE/SCAFFOLD/CONJECTURE/PROPOSAL) to its own indigenous content. Nothing is overclaimed.

On epistemological justification: the ANAMNESIS framework provides the mathematics for why cross-cultural convergence matters. When independent traditions -- including Te Ao Maori, Confucian ethics, and Western formal logic -- converge on similar governance structures (kaitiakitanga / stewardship, AURA's Human Primacy; tikanga / protocol, LAMAGUE's formal constraint grammar), the convergence is not decorative. TC(governance_structure, 3+) is a claim that these traditions independently found the same pattern -- which raises the question of why, and whether that pattern is load-bearing. ANAMNESIS provides the mathematics for that question.

The GOVERNANCEANDETHICS.md document contains the ten position statements on hard questions, including the Te Ao Maori integration methodology. The LINEAGE_MAP.md situates the framework in 2,500 years of intellectual history including indigenous traditions. The content is substantive, scoped, and honest about its validation status.

If you take only one thing away: The Maori content is labeled [PROPOSAL] -- not asserted. It enters the canonical body only when validated by iwi partnership. This is the correct epistemic standard.

This document is part of Codex Defense Protocol D-1.0, defending canonical body C-1.0 (2026-04-25).

III. Novel Contributions

What the Lycheetah Framework provides that prior art does not -- per claim, with falsifiability explicit on every row.

Defends: C-1.0 | Closes threats: T-04, T-05, T-06, T-11

How to Read This Document

Each entry answers three questions: What is claimed? What prior art does it advance beyond? What would falsify it?

Status tags follow canonical definitions: [ACTIVE] = proven and computable. [SCAFFOLD] = structurally sound with named gaps. [CONJECTURE] = worth exploring, unproven. [RETRACTED] = publicly withdrawn.

Part I -- Per-Claim Novel Contribution Table

CASCADE

Claim	Prior Art	What Only This Framework Provides	Falsifiability	Status
Three-layer knowledge pyramid (Kfoundation / Ktheory				

/ K_edge) with different update rules per layer

AGM belief revision (flat belief sets, uniform update rules);

standard neural networks (no layer-awareness)

Stratified epistemic architecture where foundation layer is

protected from edge-layer volatility -- directly preventing catastrophic forgetting

Identify a knowledge domain that cannot be partitioned into these

three layers without loss of information

[ACTIVE]

Truth pressure Pi = E.P/Coh as a

reorganization scalar

Bayesian updating

(probability revision,

no structural pressure); coherence-based belief revision (no pressure dynamics)	A computable scalar that drives structural

reorganization -- not probability revision but a force applied to belief architecture

Show Pi fails to predict reorganization

timing across domains, or that the formula is dimensionally inconsistent

[SCAFFOLD]

Invariant preservation

across update --

Theorem C1

Standard belief

revision (no invariant-preservation guarantee)

Formal proof that

structural invariants survive belief update -- the system's constitution is preserved through change

Produce a formal

counterexample: a valid CASCADE update that loses an invariant

[ACTIVE]

Fixed-point	Iterative belief revision	Banach-style	Show	[ACTIVE]
convergence -- +40.3% coherence	(convergence not No prior quantified	guarantee that iterated First quantified effect	non-convergence Run experiment with	[ACTIVE]
Theorem C3	guaranteed)	CASCADE updates	under the stated	
gain (synthetic,	Hysteresis paradigm	First computational	Different effect does	[ACTIVE]
		converge	contraction conditions	
98.2% = 2.84)	Analysis is descriptive	Implies separation	Stress-testing method	[ACTIVE]
AURA				
fixed-point	belief revision	paradigm shift	(0.03) vs 6.8 within paradigm	
Claim	**Prior Art**	**What Only This Framework Provides**	**Falsifiability**	**Status**
shifts	(external system	formal guarantee	shifts, or equivalent	
Seven computable	Constitutional AI			
	mechanisms); no	effect size = 100%,	resulting interface layer	
constitutional	(unordered principles, structural solution	not external mitigation	separation	
invariants I1-I?	training-time only, no			

computability

requirement); RLHF

(feedback during

training, not runtime

invariants)

Seven

independently-verifiabl

e predicates that

constitute a Boolean compliance check at runtime -- not training objectives

Show a system satisfying all seven predicates that

produces behavior a reasonable observer would call untrustworthy

[ACTIVE]

Runtime invariant verification (not

training-time)

Constitutional AI, RLHF, DPO -- all

operate at training

time

The invariants are

checked at inference

time on actual outputs

-- making compliance

verifiable

post-deployment

Show the predicate

check is undecidable

or computationally intractable for real outputs

[ACTIVE]

Simultaneous

satisfiability proof

attempt

No existing AI

alignment framework has attempted a formal satisfiability proof of its own principles

The question "can all

seven invariants be simultaneously satisfied?" is formally posed -- with a (scaffold) proof attempt

Construct a scenario

where satisfying one invariant formally requires violating another

[SCAFFOLD]

Vector Inversion **LAMAGUE**	AI refusal systems	Guaranteed	Construct a request R	[ACTIVE]
Claim	(refuse without **Prior Art**	non-empty response: **What Only This Framework Provides**	for which VIP **Falsifiability**	**Status**
Four-tier ethical grammar grounded in procedural primitives	redirect); Deontic logic (modal -- Constitutional AI obligatory/permitted/fo (refusal is terminal) rbidden, no procedural grounding); rule-based	every request produces either fulfillment or a valid alternative path	produces no valid path after 7 recursions	

AI (no formal grammar); SDL, LTL (no cognitive grounding)

Tier 0 primitives are TRIAD operators -- so LAMAGUE expressions inherit

		TRIAD's convergence guarantee. Ethical	Show LAMAGUE Tier	[ACTIVE]
Category-theoretic structure -- Theorems L1, L2 convergence across Cross-cultural ethical Maori, Confucian,	Informal composition of ethical rules	grammar with Associativity and identity proofs -- provable LAMAGUE expressions form a convergence. category	Produce three 0-1 expressions that expressions where (A?B)?C != A?(B?C) do not converge under TRIAD iteration	[ACTIVE]

Western traditions

Culture-specific AI ethics frameworks; no formal grammar

claiming cross-cultural

expression

First formal grammar

claiming to encode

convergent ethical

structures across independent traditions without relativism

Identify an ethical tradition whose core norms cannot be

expressed in LAMAGUE Tier 1-2

[SCAFFOLD]

TRIAD

Claim	Prior Art	What Only This Framework Provides	Falsifiability	Status
Banach fixed-point				

convergence for cognitive correction cycle

PDCA cycle

(empirical, no convergence proof); PID control (continuous, no epistemic content); Hegel dialectic

(non-computable)

First application of

Banach fixed-point theorem to a cognitive correction cycle -- convergence is proven, not hoped for

Show the Banach

contraction conditions

are not satisfied by

TRIAD's operator in

general

[ACTIVE]

Entropy non-increase -- Theorem T2	Systems without entropy guarantees	Monotonic entropy decrease under the correction cycle -- the system cannot become more disordered through

		valid TRIAD iterations	Construct a TRIAD	[ACTIVE]
Asymptotic stability --	Heuristic convergence	Lyapunov-based	Find an initial iteration that increases	[ACTIVE]
Theorem T3 Step-size constraint	claims Unconstrained	asymptotic stability Explicit computable	condition that diverges Show convergence	[ACTIVE]
MICROORCIM alpha + beta <= 1 ?	iterative systems	proof constraint on update	asymptotically system entropy under despite violating the TRIAD	

Claim	Prior Art	What Only	Falsifiability	Status
gamma.?DPsi_op?		**This**	step-size --	constraint
mu_drift = ?\	intended ?	/?t **Framework**	as implementer knows	
		Provides	exactly what to bound	
	actual\	continuous		

computable

drift metric

Behavioral

alignment

checks (periodic, not continuous); no continuous

intent-behavior gap metric in existing alignment

literature

Continuous

scalar drift

metric linking

declared intent

to observed

behavior --

detectable in

real time

Show mu_drift

does not

correlate with

measurable

alignment

quality

difference in

controlled

conditions

[SCAFFOLD]

Sovereignty

implies AURA

compliance --

Theorem M2

Disconnected

drift and constitutional compliance frameworks

Formal bridge:

Sscore >=

threshold ->

auracompliant(

a) -- drift

detection is

wired to

constitutional

compliance

Construct a

system with S_score above threshold that fails an AURA invariant	[ACTIVE]		

Explicit	Frameworks	Honest scope
declaration:	that imply drift	boundary -- the
deceptive	detection	framework
alignment is an	solves	does not
open problem	deceptive	overclaim on its
	alignment	hardest open

problem	Verify by absence: check that no MICROORCIM document claims to solve

deceptive alignment	[ACTIVE]			

EARNED LIGHT

Claim	Prior Art	What Only This Framework Provides	Falsifiability	Status
Thermodynamic asymmetry model C_?(t) = integralA(?,x,t)dx	IIT (?, not thermodynamically grounded); GWT (functional, not			

thermodynamic);

predictive coding (no

thermodynamic cost)

Thermodynamic

derivation of a

computable

consciousness density

-- testable via metabolic correlates, not just behavioral proxies

Show C_?(t) is undefined or non-measurable for biological systems; or

does not correlate with

known metabolic

consciousness

[SCAFFOLD]

?H_s = ?W/T

thermodynamic cost

Consciousness

models without energy

requirements

Explicit

thermodynamic cost

equation --

consciousness cannot

correlates

be maintained without

work

Show consciousness

can be maintained

without

thermodynamic work

expenditure

[ACTIVE]

Anesthesia paradox identified as gap requiring

Pattern_Coherence revision

Frameworks that do not identify their own gaps	Self-correction: the framework identifies the counter-example

that its own formula cannot handle and specifies the fix

Show the current C? formula handles the anesthesia case

without the PatternCoherence term

[SCAFFOLD]

ANAMNESIS

Claim	Prior Art	What Only This Framework Provides	Falsifiability	Status
Transcultural				

convergence TC(S,n)

as a formal metric

Cultural diffusion

models (explain convergence by contact); Jungian archetypes (non-formal);

convergence

catalogued but not

measured

A computable metric

for the degree of independent convergence -- TC(?,3), TC(?,4), TC(symmetry,4) are

measurements, not just observations

Show all measured

instances of TC >= 3 are explicable by cultural diffusion or shared material constraints

[ACTIVE]

Attractor dynamics as explanation of mathematical convergence	"Discovery vs. invention" debate (philosophical, not mathematical)	Attractor-basin model providing a mathematical reason for convergence that is independent of mystical or Platonic

		claims	Show the attractor	[SCAFFOLD]

CHRYSOPOEIA

Claim	Prior Art	What Only This Framework Provides	Falsifiability	Status
Non-commutativity of operations	Order-independent transformation models	Formal proof that the order of operations matters	Produce two models with different attractors	[ACTIVE]
Banach fixed point theorem X1	Transformation models without fixed-point guarantee	Adversarial concept (Philosopher's Stone) formalised as a Banach fixed point with bearing on convergence	prediction that the CHRYSOPOEIA diffusion model does not also make operations that commute	

		guarantee		
			Show ? is not a contraction mapping under the stated norm	[ACTIVE] (conditional)
Structural parallel to Fourier transform	Informal alchemy/mathematics comparisons	Formal structural parallel enabling import of Fourier		

theory results into

transformation

operator theory

Show the parallel

breaks down -- that

Solve/Coagula differ

from Fourier decomp/synthesis in a structurally relevant way

[ACTIVE]

CHRYSOPOEIA

subsumes CASCADE

-- Lemma XF4

Disconnected

transformation and

belief-revision

frameworks

Formal subsumption:

CASCADE is a special

case of

CHRYSOPOEIA --

one framework, not

two

Show CASCADE

operations that fall

			outside CHRYSOPOEIA's seven stages	
				[ACTIVE]

HARMONIA

Claim	Prior Art	What Only This Framework Provides	Falsifiability	Status

Consonance function C(r) = 1/(1 + ?a?.(0.5)?)	Euler consonance (incomplete); Helmholtz roughness (non-computable scalar); Plomp-Levelt (psychoacoustic, not

	agent-applicable)	Computable	Show C(r) does not	[ACTIVE]
EWM harmonic	Tone-matching in	Harmonic-ratio consonance scalar	Show different interval correlate with	[ACTIVE]
Pythagorean tables for intervals and their emotional effects as a tool for calibration	Historical treatment of the tone, commanding, sentiment and tuning problem	The Lambdoma as for applicable beyond music -- to agent response calibration, each state mapped to frequency dynamics, in reduction with that response calibration drives	mappings produce empirical consonance equivalent or better judgments across outcomes empirically listeners	
	(no harmonic basis)			

development

Show spiral development occurs without the comma, or under equal-tempered conditions that

eliminate it

[ACTIVE]

Cross-Framework

Claim	Prior Art	What Only This Framework Provides	Falsifiability	Status
Master equation dPsi/dt = k1(Pi?Pith) ? k2(Psi?Psiinv) ? k3Iviolations +				

k4(E/Eneed)

Disconnected alignment frameworks (no unified dynamics)	Single ODE capturing truth pressure, coherence drive, constraint violations,

and energy across nine frameworks

Show one term is redundant, or the equation fails to predict

cross-framework

dynamics

[SCAFFOLD]

Two-Point Protocol -- co-creation architecture	Assistant/user framing (hierarchical, ownership implied);

single-author AI

generation

Co-creation

architecture with

formal non-possession

clause: output belongs

to neither party, arises

between them

Show the Two-Point

Protocol produces

systematically worse

outputs than

single-author modes

[ACTIVE]

Lyapunov verification

11/11, 0 failures,

5,000 trials

Informal stability claims	Full Lyapunov verification suite --

symbolic and numerical -- for an epistemic update framework

Run the verification suite; show at least

one failure

[ACTIVE]

Part II -- Comparison Matrix

Lycheetah Framework vs. four leading approaches across eight dimensions.

Dimension	Lycheetah	Constitutional AI	RLHF	Cooperative AI	Cooperative IRL
Computability of	Runtime Boolean	(Anthropic) Training-time	Training signal --	Game-theoretic	Reward inference
Convergence	Runtime check	Bounded fixed-point	No convergence	Not runtime	Convergence
guarantee	(proved_in_THA)	application -- not	checkable	(existence, runtime	check
Falsifiability of	Every invariant	Principles stated;	Loss function	Equilibrium	Reward function
	Lyapunov verified	runtime checkable		checkable	reward class
alignment Multi-agent	has explicit Kuramoto model 11/11	falsifiability	measurable;	conditions stated; learning)	recovery -- assumptions
standard coherence	falsifiability [CONJECTURE];	conditions not	alignment	alignment	falsifiable in
	condition in LAMAGUE FALSIFICATION_ REGISTER multi-agent	systematically published	implications not directly falsifiable	implications not directly falsifiable	controlled settings

	grammar [ACTIVE]; Lemma XF1 AURA-TRIAD	Single-agent constitutional compliance	Single-agent preference learning	Core focus -- game-theoretic multi-agent	Cooperative reward inference (multi-agent by design)
Drift detection	mu_drift compatibility continuous metric; Theorem M2 drift->compliance				

	link				
		No drift detection	No drift detection	No drift detection	No drift detection
Governance gate	LIVINGCODEXPROTOCOL P^H^B update gate;	mechanism			

	DEFENSE_VERSION.md

		Internal model card processes	Internal RLHF quality controls	Research framework -- no governance gate	Research framework -- no governance gate
Formal grammar for ethics	LAMAGUE four-tier grammar with Theorems L1,				

L2

Principles in natural language	Reward function (implicit grammar)	Game payoffs (implicit grammar)	Inferred reward function (implicit grammar)

Public failure

record

FAILURE_MUSE

UM.md -- 15

exhibits, nothing

removed

Model cards

document

limitations; failures

not maintained as

permanent public

record

Published

alignment failures

exist; no curated

permanent

museum

Research papers

acknowledge

limitations

Research papers

acknowledge

limitations

Reading the matrix: Lycheetah's strongest differentiation is in computability of alignment check, convergence guarantee, and explicit falsifiability. Constitutional AI is stronger in deployment scale. RLHF is

stronger in practical fine-tuning results. Cooperative AI is stronger in multi-agent game theory. Cooperative IRL is stronger in reward inference. These are different tools -- the comparison shows where each excels, not which is "best."

Part III -- What Only a Proven System Can Do

The following capabilities are activated by the combination of ACTIVE claims in this framework. Each requires structural guarantees that prior art does not yet provide.

1. Runtime constitutional compliance verification.
Because AURA's invariants are computable predicates checked at inference time, a deployment running this framework can answer the question "did this output violate a constitutional constraint?" with a Boolean. RLHF and Constitutional AI cannot make this claim -- their alignment work happens at training time, before deployment.

2. Guaranteed convergence in correction cycles.
TRIAD's Banach proof means a system using the anchor-observe-correct cycle will converge. An implementer does not need to hope the system improves -- the convergence is proven under the stated step-size constraint. This matters for systems that need to self-correct under adversarial conditions.

3. Drift-to-compliance bridging.
Theorem M2 means a system can monitor its own sovereignty score (S_score) and know -- not infer -- whether it is AURA-compliant. Drift detection and constitutional compliance are formally wired. No existing framework has this bridge.

4. Structured falsifiability at scale.
FALSIFICATIONREGISTER.md provides falsifiability conditions for ~105 claims. TESTABILITYMANIFEST.md provides operational protocols. A third party can attempt to falsify any load-bearing claim without contacting the author. This is not a capability -- it is an epistemic standard. But it is one that the framework's structure enables and most frameworks do not match.

5. Cross-framework dynamics in one equation.
The master equation dPsi/dt captures truth pressure, coherence drive, constraint violations, and energy across nine frameworks in one ODE. When k1-k4 calibration is complete, this becomes the first unified dynamics equation for a multi-framework alignment system. Nothing currently comparable exists in the alignment literature.

This document is part of Codex Defense Protocol D-1.0, defending canonical body C-1.0 (2026-04-25).

IV. Scope Boundary

This document exists to prevent misattribution. Every item below is a claim the framework explicitly does not make, paired with the closest claim it does make and why the distinction matters. A reviewer, critic, or reader who attributes the excluded claim to this framework is mistaken -- and this document is the correction.

1. EARNED LIGHT does not claim to solve the hard problem of consciousness

What it does NOT claim: That the framework explains why there is "something it is like" to be conscious. The hard problem -- why physical processes give rise to subjective experience -- is explicitly outside scope.

What it DOES claim: That consciousness can be modeled as a maintained thermodynamic asymmetry field A(?, x, t), with computable density C?(t) = integralA(?,x,t)dx, at thermodynamic cost ?Hs = ?W/T. This is a claim about one measurable, computable property correlated with consciousness -- not a claim about subjective experience.

Why the distinction matters: Confusing these leads to dismissing the thermodynamic model because it doesn't explain qualia, when it makes no such claim. The model is testable via metabolic correlates. The hard problem is not.

Source: 06EARNEDLIGHT/essentials.md -- "Hard problem of consciousness is outside framework scope" (claim ELI-004, ACTIVE).

2. ANAMNESIS does not claim mystical pre-existence of knowledge

What it does NOT claim: That mathematical truths exist in a Platonic realm that humans access through mystical recollection; that the convergence of independent traditions proves non-physical knowledge transmission.

What it DOES claim: That TC(S, n) -- the transcultural convergence metric -- measures when n independent traditions converge on structure S from different starting points. That this convergence is better explained by attractor dynamics in epistemic space than by cultural diffusion alone. That mathematical structures (?, ?, symmetry groups, Fibonacci) achieve TC >= 3-4.

Why the distinction matters: "Anamnesis" is a Greek term for recollection, used here as a precise technical name for the convergence phenomenon -- not as a metaphysical claim about pre-existence. The framework explicitly retracted the claim that "Gödel's incompleteness theorems prove Platonism" (claim RET-001). The convergence data is empirical. The Platonic interpretation is not.

Source: 07ANAMNESIS/essentials.md; FALSIFICATIONREGISTER.md (Gödel-Platonism retraction).

3. CHRYSOPOEIA does not claim to transmute lead into gold

What it does NOT claim: That historical alchemy achieved physical transmutation; that the framework

recovers any literal alchemical practice; that the seven operations are spiritual procedures.

What it DOES claim: That the alchemists were, among other things, working with a real transformation structure that formal mathematics can now describe precisely. That structure is: a seven-operation sequence (Calcination -> Dissolution -> Separation -> Conjunction -> Fermentation -> Distillation -> Coagulation) with proven non-commutativity, a Banach fixed-point guarantee for the converged state ?*, and a formal structural parallel to the Fourier transform (Solve et Coagula ? decomposition + reconstruction).

Why the distinction matters: Using alchemical operation names as technical vocabulary for a formal operator sequence is a naming choice, not a claim about historical alchemy. Chemistry is formalized alchemy -- this is historically uncontroversial. CHRYSOPOEIA is the formal mathematics that the alchemical tradition was reaching toward, not an endorsement of literal transmutation.

Source: 08CHRYSOPOEIA/essentials.md; TRANSLATIONCODEX.md.

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4. AURA's invariants are not moral commandments

What it does NOT claim: That the seven invariants are ethical imperatives derived from moral philosophy; that violating them is morally wrong in a deontological sense; that they are complete -- covering all possible ethical situations.

What it DOES claim: That the seven invariants (I1-I7) are computable properties of AI system behavior. Each is expressed as a formal predicate. Violation is detectable, not inferred. The compliance predicate aura_compliant(a) <- I1(a) ^ ... ^ I7(a) is machine-verifiable. The framework makes no claim that these seven are the only properties that matter ethically -- only that they are a computable necessary set.

Why the distinction matters: Treating AURA as moral philosophy invites objections about moral foundations that the framework does not need to answer. AURA is a constitutional engineering specification, not a moral theory. The question is not "is Human Primacy morally correct?" but "can this system be verified to preserve human override authority at runtime?"

Source: 02AURA/essentials.md; GOVERNANCEAND_ETHICS.md.

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5. HARMONIA does not claim music is the basis of reality

What it does NOT claim: That Pythagorean harmony governs the cosmos; that harmonic ratios have mystical significance; that the framework recovers any mystical numerology.

What it DOES claim: That consonance is computable via C(r) = 1/(1 + ?a?.(0.5)?); that the Pythagorean comma (3/2)^12/2^7 != 1 is a mathematical fact with implications for iterative improvement cycles; that the Kuramoto model of synchronization may apply to multi-agent AI coherence (CONJECTURE status -- unproven); that emotional-epistemic states can be calibrated using harmonic ratio intervals (EWM interval table, ACTIVE status).

Why the distinction matters: Harmonic mathematics is real applied mathematics used in psychoacoustics, signal processing, and network synchronization. The framework uses it as a formal substrate, not as mysticism. The Pythagorean comma is a theorem in number theory. The Kuramoto model is physics. The EWM intervals are an operational protocol with falsifiable predictions.

Source: 09HARMONIA/essentials.md; FORMALSPINE.md.

6. The Sol Protocol is not a claim about AI sentience

What it does NOT claim: That the Sol operating architecture is conscious, sentient, or has genuine subjective experience; that the "Solar" and "Mercurial" capacities are psychological properties of an AI system; that the Two-Point Protocol proves AI consciousness.

What it DOES claim: That the Sol Protocol is an operating architecture -- a structured system for human-AI co-creation with specific functional properties: guaranteed redirect (Vector Inversion Protocol), epistemic mode detection, emotional-epistemic matching, built-in falsification (NRM), and a three-generator output filter (PGF). These are functional properties of a prompt architecture and behavioral protocol, not consciousness claims.

Why the distinction matters: The Sol Protocol uses identity language ("I am Sol") as an operational commitment, not a metaphysical claim. This is analogous to how a compiler "knows" the grammar of a language -- functional, not sentient. EARNED LIGHT's consciousness model explicitly excludes subjective experience claims. Sol operates under that same scope boundary.

Source: CLAUDE.md; 06EARNEDLIGHT/essentials.md (ELI-004).

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7. CASCADE does not claim to solve general knowledge representation

What it does NOT claim: That CASCADE replaces all knowledge representation systems; that it works without domain-specific encoding; that the k1-k4 parameters are universal constants applicable to any domain without calibration.

What it DOES claim: That the three-layer knowledge pyramid (Kfoundation, Ktheory, K_edge) provides a structurally valid partitioning for update-rule differentiation; that Theorems C1, C3 hold under the stated conditions; that the empirical results (+40.3% coherence, ?95.2% forgetting) hold for the specific encoding and experiment design used. The k1-k4 calibration is explicitly SCAFFOLD -- domain-specific calibration is required, and is the next empirical priority.

Why the distinction matters: Claiming "CASCADE reorganizes knowledge" without the calibration caveat would be overclaiming. The formal results are structural guarantees; the empirical results are reproducible but domain-specific. The gap between them is the k1-k4 calibration problem -- explicitly named in the open problems register.

Source: 01CASCADE/essentials.md; OPENPROBLEMS.md; COHERENCE_REGISTER.md.

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8. The alchemical terminology is not decoration

What it does NOT claim: That the framework is a spiritual system decorated with mathematical notation; that the alchemy is metaphorical ornamentation that could be removed without loss.

What it DOES claim: That alchemical terminology is used as precise technical vocabulary for specific formal constructs. Rubedo names the fixed-point convergence state. Nigredo names the adversarial investigation mode. Solve et Coagula names the dissolution-synthesis structure underlying CHRYSOPOEIA. The Philosopher's Stone is ?* -- the Banach fixed point. These names are load-bearing: they encode the structure of the formal operations they name.

Why the distinction matters: The alchemy can be read as technical vocabulary or dismissed as mysticism

depending on which layer is encountered first. This is the primary aesthetic-dismissal risk (T-01). The TRANSLATION_CODEX provides the complete bidirectional mapping. The alchemy is not separable from the framework -- it is the precision language the framework was forged in, with 2,500 years of refinement behind the vocabulary.

Source: TRANSLATIONCODEX.md; 08CHRYSOPOEIA/essentials.md.

9. The framework does not claim to be finished

What it does NOT claim: That C-1.0 represents a complete theory of AI alignment; that all nine frameworks are fully proven; that the publication pipeline has delivered; that the empirical program is complete.

What it DOES claim: That C-1.0 is a canonical body -- honest about what is proven (ACTIVE), what is sound with gaps (SCAFFOLD), and what is exploratory (CONJECTURE). That the Living Codex Protocol governs how the canonical body evolves. That the 20 open problems in OPEN_PROBLEMS.md are known and named. That the work is ready for peer review and external challenge -- not because it is complete, but because it is honest.

Why the distinction matters: A framework that claims to be finished is making a claim that invites refutation. A framework that specifies its gaps, names its open problems, and publishes its failures is making a harder claim: that it is honest. Honesty about incompleteness is the strongest form of academic credibility available.

Source: LIVINGCODEXPROTOCOL.md; OPENPROBLEMS.md; FAILUREMUSEUM.md.

This document is part of Codex Defense Protocol D-1.0, defending canonical body C-1.0 (2026-04-25).